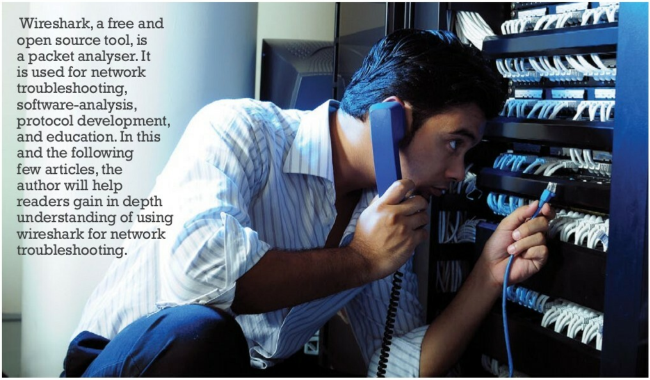
Wireshark: Essential for a Network Professional's Toolbox



Wireshark, a free and open source tool, is a packet analyser. It is used for network troubleshooting, software-analysis, protocol development, and education. In this and the following few articles, the author will help readers gain in depth understanding of using wireshark for network troubleshooting.

In order to troubleshoot computer network related problems effectively and efficiently, an in-depth understanding of TCP/IP is absolutely necessary, but along with that, you also need to 'see' TCP/IP traffic. There are various open source and commercial tools available to capture and display network traffic for further analysis.

Let us consider the essential features of effective traffic analysis and network troubleshooting tools:

- The ability to capture traffic from various networks such as wired, wireless, etc
- Layer-wise representation of captured traffic for easy readability
- Ability to save the packets in standard format
- Ability to drill down packets for traffic analysis

All these features are an inherent part of Wireshark. Over and above this, the tool is available under GNU GPL (read, free). So Wireshark is an absolute must for any networking professional!

This series of articles on Wireshark will familiarise readers with the Wireshark GUI and analysing various TCP/IP protocols by means of captured packets, explaining the features of Wireshark and discuss various scenarios to locate network related problems.

The basics: Wireshark modules

To provide the desired functions, Wireshark uses a number of different modules integrated together by the Wireshark core.

Module	Function
GTK (GIMP Toolkit)	GUI and handling of all user inputs/ outputs
Dumpcap	Packet capture engine
Pcap libraries	Actual packet capturing. The Linux version uses *libpcap* developed by the Tcpdump team (available under the BSD licence), while the Windows version uses *winpcap* developed by Riverbed Technologies, which is available as freeware
Dissector(s)	Detects the packet's type and gets the maximum information out of it – you can also write custom dissectors
Wiretap libraries	Reading and writing captured files from and to various file formats

Here, I have taken the liberty of mentioning only the important or relevant modules and their functions for easy understanding. For more details, please refer to *wireshark.org*.